JEMIMA RAHMAN APPI

Computer Science Student · ML & Backend Developer

Contact

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Education

Bachelor of Science in Computer Science and Engineering, Jahangirnagar University

Skills

Programming Languages

- Python, C, C++, Java, SQL

Backend Development

- FastAPI, REST APIs
- SQLAlchemy, Alembic
- JWT Authentication
- Pytest

Web Development

- HTML, CSS, JavaScript, Bootstrap, PHP

Databases

- PostgreSQL, MySQL, SQLite

Machine Learning & Data Science

- Scikit-learn, TensorFlow
- XGBoost, Prophet, SHAP
- Pandas, NumPy, EDA
- Regression, SVM, Decision Trees, Random Forest, KNN, K-Means
- ANN, RNN, LSTM

AI / LLM & MLOps

- LangChain, Pydantic
- Streamlit
- MLflow, Evidently AI

Tools

- Git, GitHub, VS Code, Postman, Google Colab, MATLAB

ABOUT ME

Final-year Computer Science and Engineering student with hands-on experience in Machine Learning, LLM applications and backend development with FastAPI. Seeking opportunities to apply technical skills and contribute to innovative software solutions.

Experience

- **Data Science & Analytics Intern** – ***View Certificate***
  Zidio Development Pvt. Ltd. (Remote) · Jul 2026 – Aug 2026
  - Contributed to Data Science and Analytics project work; performance recognized as commendable in the experience certificate.

Projects

- **RetailPulse – AI Retail Analytics**
  Python, Prophet, XGBoost, SHAP, Scikit-learn, Streamlit, MLflow, Evidently AI
  - Built demand forecasting (Prophet), RFM + K-Means customer segmentation and XGBoost churn prediction with SHAP explainability.
  - Added inventory reorder recommendations, data drift monitoring and MLflow experiment tracking; deployed as a Streamlit dashboard.
- **Study Buddy Chatbot**
  Python, LangChain, Groq LLM, Pydantic, Streamlit
  - AI study assistant that classifies questions by subject and routes them to subject-specific chains using RunnableBranch.
  - Generates answer, summary and keywords in parallel (RunnableParallel) as Pydantic structured output.
- **Personal Expense Tracker API**
  Python, FastAPI, SQLAlchemy, PostgreSQL, JWT, Pytest
  - REST API with JWT authentication, bcrypt password hashing and per-user income/expense transactions.
  - Filtering by type, category and amount range, with automated tests using Pytest.
- **ToDo App API**
  Python, FastAPI, SQLAlchemy, Alembic, SQLite, JWT
  - Task management API with JWT login, role-based admin access, priorities and Alembic database migrations.
- **Social Media Ads Prediction**
  Python, Scikit-learn, Pandas, NumPy
  - Machine learning model predicting customer purchase behavior from social media ads using classification algorithms.
- **Smart Exam Portal** – ***PHP, MySQL, Bootstrap***
  Online exam system with student login, result generation and performance tracking.
- **Graphing Calculator** – ***Java***
  Evaluates math expressions and plots functions as dynamic graphs.

Certifications

freeCodeCamp – Responsive Web Design Certification